Objective:

The goal of this assignment is to test a simple web application and perform both manual and automated testing to evaluate software quality. You should apply Software Quality Assurance (SQA) techniques, document the testing process, and demonstrate the results during a viva session.

Group Formation & Submission Guidelines:

- Each group should consist of four members.
- The final report and test cases document must be submitted **on or before the 17th April**.
- The viva session schedule will be announced later.

Assignment Tasks and Deliverables:

1. System specification

- Each group must test the given system that will serve as the test subject for this assignment.

  System URL– https://www.pixelssuite.com/

  Testing can be performed with any other feature except for 'Transliteration' feature in the given system link.

- Ensure that the application is tested considering its functionality and use both manual and automated testing.

2. Testing Process

A. Manual Testing

- Identify different test scenarios, including functional testing, UI testing, and edge cases.
- Perform manual testing on the developed application by designing and executing test cases.

Test Case Document should consist of the following details,

- Test case ID
- Test scenario
- Test steps

- Expected results
- Actual results
- Status (Pass/Fail)
- Document the results for each test case with actual results.
- Report any defects or issues found during testing.

B. Automated Testing

- Select preferred test automation tools (Playwright, Selenium, TestNG, Cypress, etc.) to perform automate testing.
- Each member should **use a separate automation tool** to perform testing, ***where usage of Playwright is mandatory for one member***. The other three tools for the remaining members can be of any choice.
- Develop automated test scripts to cover essential functionalities of the application.
- Execute automated tests and document the results.
- Capture screenshots of test executions and attach the automation scripts to the test cases document.

3. Report Submission

- Ensure that your *final report* does not exceed three pages, with additional documentation, including any screenshots, attached to the **Appendix** section.
- Test cases should be included in a separate detailed document.
- The report and the test cases document must be submitted as a PDF file by the specified deadline. They should be named in below format.

  <Leader's_ITNumber>_SQA_final_report_2026

  <Leader's_ITNumber>_SQA_test_cases_document_2026

  The submission should be made by submitting the documents as a single zip folder with the name *<Leader's_ITNumber>_SQA_assignment_submission.*
- Each group must submit the comprehensive report and the test cases document that includes the following sections:

1.Final report

I. Introduction (Brief introduction covering following points)

- Overview of the system tested
- Purpose and scope of testing
- Test Plan
- Objectives of testing.
- Testing methodologies used (manual and automated).
- Tools and technologies used.

II. Manual Testing Overview

- Overview of manual testing.
- Defects found.

III. Automated Testing Overview

- Selected automation tool and reasoning for its selection.
- Overview of automated testing.
- Defects found.

IV. List of identified defects with priority and severity levels.

V. Conclusion

- Summary of findings.
- Comparison on the testing tools used.
- Challenges faced and improvements suggested.

References

Appendix

2.Test cases document

1. **Manual Testing Documentation**

- Test cases document including Test case ID, Test scenario, Test steps, Expected results, Actual results and Status (Pass/Fail)
- Screenshots of manual test executions.

2. **Automated Testing Documentation**

- Automated test cases with scripts.
- Test execution results with screenshots.
- Defects found.

4. Viva Session

- Each group member should participate in viva session where they will present their testing process and results.
- During the viva, the group must:
  - Demonstrate at least two automated test cases using the selected automation tool.
  - Show testing outputs, including manual and automated test results.
  - Answer questions related to the testing process and outcomes.
  - All four group members must actively participate in the viva session and **should be aware of all the tools used in testing**, and each member will be evaluated individually.

Assignment Marking Guide
Total Marks – 20 Marks

Criteria	**Description**	**Marks Allocated**
Test Plan	A well-structured test plan covering scope, objectives, test methodologies, tools, and execution strategy.	3 marks
Manual Test Cases and Execution	Well-documented test cases with clear steps, expected results, actual results, and pass/fail status. Screenshots included.	4 marks
Automation Execution and Results	Successfully executes automation scripts with screenshots and reports demonstrating results.	3 marks
Report Quality and Documentation	The report is well-structured, detailed, and includes all required sections with clarity.	4 marks
Viva and Demonstration	Clear and confident presentation, demonstrating two automated test cases successfully, with active participation.	4 marks
Team collaboration	Demonstrate equal contribution and all members should be aware of the complete testing process of the system.	2 marks